Exercises

Level - 1

(Problems Based on Fundamentals)

ABC OF CIRCLES

1. Find the centre and the radius of the circle

(i) $x^2 + y^2 = 16$

(ii) $x^2 + y^2 - 8x + 15 = 0$

(iii) $x^2 + y^2 - x - y = 0$

2. Prove that the radii of the circles

$x^2 + y^2 = 1$, $x^2 + y^2 - 2x - 6xy = 6$.

and $x^2 + y^2 - 4x - 12y = 9$ are in AP.

3. Find the equation of the circle contric with the circle $x^2 + y^2 - 8x + 6y - 5 = 0$ and passing through the point $(-2, -7)$.

4. Find the equation of the circle passing through the point of intersection of $x + 3y = 0$ and $2x - 7y = 0$ and whose centre is the point of intersection of lines $x + y + 1 = 0$ and $x - 2y + 4 = 0$.

5. Find the equation of the circle touching the lines $4x - 3y = 30$ and $4x - 3y + 10 = 0$ having the centre on the line $2x + y = 0$.

6. Let the equation of a circle is $x^2 + y^2 - 16x - 24y + 183 = 0$. Find the equation of the image of this circle by the line mirror $4x + 7y + 13 = 0$.

7. Find the area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

8. A circle has radius 3 units and its centre lies on the line $y = x - 1$. Find the equation of the circle if it passes through $(7, 3)$.

9. Find the point P on the circle $x^2 + y^2 - 4x - 6y + 9 = 0$ such that $\angle POX$ is minimum, where O is the origin and OX is the x-axis.

10. Find the equation of the circle when the end-points of a diameter are $(2, 3)$ and $(6, 9)$.

11. Find the equation of a circle passing through $(1, 2)$, $(4, 5)$ and $(0, 9)$.

12. Find the equation of a circle passing through the points $(1, 2)$ and $(3, 4)$ and touching the line $3x + y = 5$.

13. Find the length of intercepts to the circle $x^2 + y^2 + 6x + 10y + 8 = 0$.

14. Find the length of y-intercept to the circle $x^2 + y^2 - x - y = 0$.

15. Show that the circle $x^2 + y^2 - 2ax + a^2 = 0$ touches both the co-ordinate axes.

POSITION OF A POINT WITH RESPECT TO A CIRCLE

16. Discuss the position of the points $(1, 2)$ and $(6, 0)$ with respect to the circle $x^2 + y^2 - 4x + 2y - 11 = 10$.

17. If the point $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then find range of λ.

SHORTEST AND LONGEST DISTANCE OF A CIRCLE FROM A POINT

18. Find the shortest and the longest distance from the point $(2, -7)$ to the circle

$x^2 + y^2 - 14x - 10y - 151 = 0$

INTERSECTION OF A LINE AND A CIRCLE

19. Prove that the line $y = x + 2$ touches the circle $x^2 + y^2 = 2$. Find its point of contact.

20. Find the equation of the tangent to the circle $x^2 + y^2 = 4$ parallel to the line $3x + 2y + 5 = 0$.

21. Find the equation of the tangent to the circle $x^2 + y^2 = 9$.

22. Find the equation of the tangent to the circle $x^2 + y^2 + 4x + 3 = 0$, which makes an angle of 60° with the positive direction of x-axis.

23. If a tangent is equally inclined with the co-ordinate axes to the circle $x^2 + y^2 = 4$, find its equation.

24. Find the equation of the tangents to the circle $x^2 + y^2 = 25$ through $(7, 1)$.

TANGENT TO A CIRCLE

25. If the centre of a circle $x^2 + y^2 = 9$ is translated 2 units parallel to the line $x + y = 4$ where x increases, find its equation.
26. Find the equation of the tangents to the circle $x^2 + y^2 = 9$ at $x = 2$.
27. Find the equation of the tangent to the circle $x^2 + y^2 = 16$ at $y = 4$.
28. Find the points of intersection of the line $4x - 3y = 10$ and the circle.
29. Find the equation of the pair of tangents drawn to the circle $x^2 + y^2 - 2x + 4y = 0$ from (0, 1).
30. Find the equation of the common tangent to the curves $x^2 + y^2 = 4$ and $y^2 = 4(x - 2)$.
31. Find the shortest distance between the circle $x^2 + y^2 = 9$ and the line $y = x - 8$.

LENGTH OF THE TANGENT TO A CIRCLE

32. Find the length of the tangent from the point (2, 3) to the circle $x^2 + y^2 = 4$.
33. Find the length of the tangent from any point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$.
34. Find the length of the tangent from any point on the circle $x^2 + y^2 + 2011x + 2012y + 2013 = 0$ to the circle $x^2 + y^2 + 2011x + 2012y + 2014 = 0$.

POWER OF A POINT W.R.T A CIRCLE

35. Find the power of a point (2, 5) with respect to the circle $x^2 + y^2 = 16$.
36. If a point $P(1, 2)$ is rotated about the origin in an anti-clockwise sense through an angle of 90° say at Q, then find the power of a point Q with respect to the circle $x^2 + y^2 = 4$.

PAIR OF TANGENTS

37. Find the equation of the tangent from the point (1, 2) to the circle $x^2 + y^2 = 4$.
38. Find the equation of the tangent to the circle $x^2 + y^2 - 4x + 3 = 0$ from the point (2, 3).
39. Find the angle between the tangent from the point (3, 5) to the circle $x^2 + y^2 = 25$.
40. If a point (1, 2) is translated 2 units through the positive direction of x-axis and then tangents drawn from that point to the circle $x^2 + y^2 = 9$, find the angle between the tangents.

DIRECTOR CIRCLE

41. Find the locus of the point of intersection of two perpendicular tangents to a circle $x^2 + y^2 = 25$.
42. Tangents are drawn from an arbitrary point on the line $y = x + 1$ to the circle $x^2 + y^2 = 9$. If those tangents are orthogonal to each other, find the locus of that point.
43. Tangents are drawn from any point on the circle $x^2 + y^2 = 20$ to the circle $x^2 + y^2 = 10$. Find the angle between their tangents.
44. Find the equation of the director circle to each of the following given circles.
    (i) $x^2 + y^2 + 2x = 0$
    (ii) $x^2 + y^2 + 10y + 24 = 0$
    (iii) $x^2 + y^2 + 16x + 12y + 99 = 0$
    (iv) $x^2 + y^2 + 2gx + 2fy + c = 0$.
    (v) $x^2 + y^2 - ax - by = 0$.

NORMAL AND NORMALCY

45. Find the equation of the normal to the circle $x^2 + y^2 = 9$ at $x = 2$.
46. Find the equation of the normal to the circle $x^2 + y^2 + 2x + 4y + 4 = 0$ at (–2, 1).
47. Find the equation of a normal to a circle $x^2 + y^2 - 4x - 6y + 4 = 0$, which is parallel to the line $y = x - 3$.
48. Find the equation of the normal to a circle $x^2 + y^2 - 8x - 12y + 99 = 0$, which is perpendicular to the line $2x - 3y + 10 = 0$.

CHORD OF CONTACT

49. Find the equation of the chord of contact of the tangents drawn from (5, 3) to the circle $x^2 + y^2 = 25$.
50. Find the co-ordinates of the point of intersection of the tangents at the points where the line $2x + y + 12 = 0$ meets the circle $x^2 + y^2 - 4x + 3y - 1 = 0$.
51. Find the condition that the chord of contact from an external point (h, k) to the circle $x^2 + y^2 = a^2$ subtends a right angle at the centre.
52. Tangents are drawn from the point (h, k) to the circle $x^2 + y^2 = a^2$. Prove that the area of the triangle formed by the tangents and their chord of contact is $a\left(\dfrac{(h^2 + k^2 - a^2)^{3/2}}{(h^2 + k^2)}\right)$.
53. The chord of contact of tangents drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$ such that $b^m = a^n c^p$, where $m, n, p \in N$, find the value of $m + n + p + 10$.

CHORD BISECTED AT A POINT

54. Find the equation of the chord of the circle $x^2 + y^2 = 25$, which is bisected at the point (–2, 3).
55. Find the equation of the chord of the circle $x^2 + y^2 + 6x + 8y - 11 = 0$, whose mid-point is (1, –1).
56. Find the locus of the middle points of the chords of the circle $x^2 + y^2 = a^2$ which pass through the fixed point (h, k).
57. Find the locus of the mid-point of a chord of the circle $x^2 + y^2 = 4$ which subtend a right angle at the centre.
58. Let AB be a chord of the circle $x^2 + y^2 = 4$ such that $A = (\sqrt{3}, 1)$. If the chord AB makes an angle of 90° about the origin in anti-clockwise direction, then find the co-ordinates of B.
59. Let CD be a chord of the circle $x^2 + y^2 = 9$ such that $C = (2\sqrt{2}, 1)$. If the chord CD makes an angle of 60°

about the centre of the circle in clockwise direction, find the co-ordinates of D.

60. Find the locus of the mid-points of the chords of the circle $x^2 + y^2 = 9$ which are parallel to the line $2012x + 2013y + 2014 = 0$.
61. Find the locus of the mid-points of the chords of the circle $x^2 + y^2 - 4x - 6y = 0$, which are perpendicular to the line $4x + 5y + 10 = 0$.

COMMON CHORD OF TWO CIRCLES

62. Find the lengths of the common chord of the circles $x^2 + y^2 + 3x + 5y + 4 = 0$ and $x^2 + y^2 + 5x + 3y + 4 = 0$.
63. Find the equation of the circle whose diameter is the common chord of the circles
$$x^2 + y^2 + 2x + 3y + 1 = 0$$
and $x^2 + y^2 + 4x + 3y + 2 = 0$.

INTERSECTION OF TWO CIRCLES

64. Find the number of tangents between the given circles
    (i) $x^2 + y^2 = 4$ and $x^2 + y^2 - 2x = 0$
    (ii) $x^2 + y^2 + 4x + 6y + 12 = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$
    (iii) $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$
    (iv) $x^2 + y^2 - 4x - 4y = 0$ and $x^2 + y^2 + 2x + 2y = 0$
    (v) $x^2 + y^2 = 64$ and $x^2 + y^2 - 4x - 4y + 4 = 0$
    (vi) $x^2 + y^2 - 2(1 + \sqrt{2})x - 2y + 1 = 0$ and $x^2 + y^2 - 2x - 2y + 1 = 0$.
65. Find all the common tangents to the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ and $x^2 + y^2 + 6x - 2y + 1 = 0$.
66. If two circles $x^2 + y^2 + c^2 = 2$ and $x^2 + y^2 + c^2 = 2by$ touches each other externally, prove that $\frac{1}{a^2} = \frac{1}{b^2} + \frac{1}{c^2}$.
67. If two circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 8x - 6y + n^2 = 0$, where n is any integer, have exactly two common tangents, find the number of possible values of n.

ORTHOGONAL CIRCLES

68. Find the angle at which the circles $x^2 + y^2 + x + y = 0$ and $x^2 + y^2 + x - y = 0$ intersect.
69. If the circles $x^2 + y^2 + 2a_1x + 2b_1y + c_1 = 0$ and $x^2 + y^2 + a_2x + b_2y + c_2 = 0$ intersect orthogonally, prove that $a_1a_2 + b_1b_2 = c_1 + c_2$.
70. If two circles $2x^2 + 2y^2 - 3x + 6y + k = 0$ and $x^2 + y^2 - 4x + 10y + 16 = 0$ cut orthogonally, find the value of k.
71. A circle passes through the origin and centre lies on the line $y = x$. If it cuts the circle $x^2 + y^2 - 4x - 6y + 18 = 0$ orthogonally, find its equation.
72. Find the locus of the centres of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 5x + 4y - 2 = 0$ orthogonally.
73. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally, find the locus of its centre.
74. Two circles having radii r_1 and r_2 intersect orthogonally, find the length of the common chord.

RADICAL AXIS

75. Find the radical axis of the two circles
$$x^2 + y^2 + 4x + 6y + 9 = 0$$
and $x^2 + y^2 + 3x + 8y + 10 = 0$
76. Find the image of a point (2, 3) with respect to the radical axis of two circles $x^2 + y^2 + 8x + 2y + 10 = 0$ and $x^2 + y^2 - 2x - y - 8 = 0$.
77. Find the radical centre of the three circles $x^2 + y^2 = 1$, $x^2 + y^2 - 8x + 15 = 0$ and $x^2 + y^2 + 10y + 24 = 0$.
78. Find the equation of a circle which cuts orthogonally every member of the circles
$$x^2 + y^2 - 3x - 6y + 14 = 0,$$
$$x^2 + y^2 - x - 4y + 8 = 0$$
and $x^2 + y^2 + 2x - 6y + 9 = 0$
79. Find the equation of the circle passing through the points of intersection of the circles
$$x^2 + y^2 + 13x - 3y = 0$$
and $2x^2 + 2y^2 + 4x - 7y - 25 = 0$
and the point (1, 1).
80. If the circle $x^2 + y^2 + 2x + 3y + 1 = 0$ cuts the circle $x^2 + y^2 + 4x + 3y + 2 = 0$ in two points, say A and B, find the equation of the circle as AB as a diameter.
81. Find the equation of the smallest circle passing through the intersection of the line $x + y = 1$ and the circle $x^2 + y^2 = 9$.
82. Find the equation of the circle, which is passing through the point of intersection of the circles
$$x^2 + y^2 - 6x + 2y + 4 = 0$$
and $x^2 + y^2 + 2x - 4y - 6 = 0$
and its centre lies on the line $y = x$.
83. Find the circle whose diameter is the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$.

Hints & Solutions

Level - 1

1. (i) The given equation of circle is $x^2 + y^2 = 16$.
Hence, the centre is (0, 0) and the radius = 4.

(ii) The given equation of a circle is
$x^2 + y^2 - 8x + 15 = 0$.
$\Rightarrow$ $(x - 4)^2 + y^2 = 16 - 15 = 1$
Hence, the centre is (4, 0) and the radius = 1.

(iii) The given equation of a circle is
$x^2 + y^2 - x - y = 0$

$$\Rightarrow \left(x-\frac{1}{2}\right)^2+\left(y-\frac{1}{2}\right)^2=\frac{1}{4}+\frac{1}{4}=\frac{1}{2}=\left(\frac{1}{\sqrt{2}}\right)^2$$

the centre is $\left(\frac{1}{2},\frac{1}{2}\right)$ and the radius is $\frac{1}{\sqrt{2}}$.

2. The equations of given circles are
$x^2 + y^2 = 1$ ...(i)
$x^2 + y^2 - 2x - 6y = 6$ ...(ii)
and
$x^2 + y^2 - 4x\ 12y = 9$ ...(iii)
Let, r_1, r_2 and r_3 are the radii of the circles (i), (ii) and (iii).
Then $r_1 = 1$,

$$r_2=\sqrt{g^2+f^2-c}=\sqrt{1+9+6}=4$$

and $r_3=\sqrt{4+36+9}=7$
Therefore, $r_1 + r_3 = 7 + 1 = 8 = 2(4) = 2r_2$.
Thus, r_1, r_2 are in AP.

3. Since the circle is concentric, so the centre of the circle is the same as
$x^2 + y^2 - 8x + 6y - 5 = 0$
Let CP is the radius.

Then $CP=\sqrt{(-4-2)^2+(-7+3)^2}$
$=\sqrt{36+16}=\sqrt{52}=2\sqrt{13}$

Hence, the equation of the circle is
$(x - 4)^2 + (y + 3)^2 = (2\sqrt{13})^2$
$\Rightarrow$ $x^2 + y^2 - 8x + 6y = 27$

4. The point of intersection of $x + 3y = 0$ and $2x - 7y = 0$ is P(0, 0) and the point of intersection of $x + y + 1 = 0$
and
$x - 2y + 4 = 0$ is C(–2, 1).
Thus, CP is the radius, i.e. $CP = \sqrt{4+1}=\sqrt{5}$
Hence, the equation of the circle is
$(x + 2)^2 + (y - 1)^2 = (\sqrt{5})^2$
$\Rightarrow$ $x^2 + y^2 + 4x - 2y = 0$

5. Clearly, the radius of the circle is

$$=\frac{1}{2}\left|\frac{30-(-10)}{\sqrt{4^2+3^2}}\right|=\frac{1}{2}\times\frac{40}{5}=4$$

Let (h, k) be the centre of the circle.
Thus $2h + k = 0$

$\Rightarrow \quad k = -2h$

Also, $\left|\dfrac{8h - 3k - 30}{\sqrt{4^2 + 3^2}}\right| = 4$

$\Rightarrow \quad \left|\dfrac{8h - 3(-2h) - 30}{5}\right| = 4$

$\Rightarrow \quad \left|\dfrac{14h - 30}{5}\right| = 4$

$\Rightarrow \quad 14h - 30 = \pm 20$

$\Rightarrow \quad 14h = 30 \pm 20 = 50, 10$

$\Rightarrow \quad h = \dfrac{25}{7}, \dfrac{5}{7}$

So, $\quad k = -\dfrac{50}{7}, -\dfrac{10}{7}$

Hence, the equation of the circle is

$$\left(x - \frac{25}{7}\right)^2 + \left(y + \frac{50}{7}\right)^2 = 16$$

or $$\left(x - \frac{5}{7}\right)^2 + \left(y + \frac{10}{7}\right)^2 = 16$$

6. The centre of the circle

$x^2 + y^2 - 16x - 24y + 183 = 0$ is $C(8, 12)$.

Let the centre of the new circle be $C'(h, k)$.

Now, $\dfrac{h - 8}{-4} = \dfrac{k - 12}{7} = -\dfrac{2(-32 + 84 + 13)}{16 + 49}$

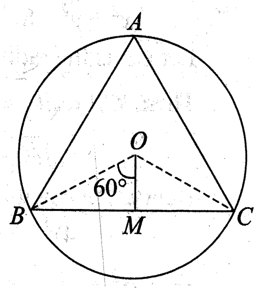
$\Rightarrow \quad h = 0, k = -2$

Hence, the equation of the new circle is

$x^2 + (y + 2)^2 = (3\sqrt{3})^2$

$\Rightarrow \quad x^2 + Y^2 + 4Y = 23$

7. Let ABC be an equilateral triangle such that OM is perpendicular on BC.



Here, $OB = \sqrt{g^2 + f^2 - c}$

Clearly, $\angle BOM = 60°$.

Now, $\sin\left(\dfrac{\pi}{3}\right) = \dfrac{BM}{OB}$

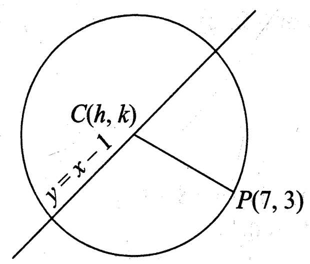
$\Rightarrow \quad BM = \dfrac{\sqrt{3}}{2} OB = \dfrac{\sqrt{3}}{2} \times \sqrt{g^2 + f^2 - c}$

Thus, $ar\ (\Delta ABC) = \dfrac{\sqrt{3}}{4} \times (BC)^2$

$= \dfrac{\sqrt{3}}{4} \times (2BM)^2$

$= \dfrac{3\sqrt{3}}{4} \times (g^2 + f^2 - c)$

8. Let the centre of the circle be (h, k) such that $k = h - 1$.



We have,

$(h - 7)^2 + (k - 3)^2 = 3^2$

$\Rightarrow \quad (h - 7)^2 + (h - 4)^2 = 3^2$

$\Rightarrow \quad h^2 - 11h + 28 = 0$

$h = 7, 4$ and $k = 6, 3$

Hence, the equation of a circle can be

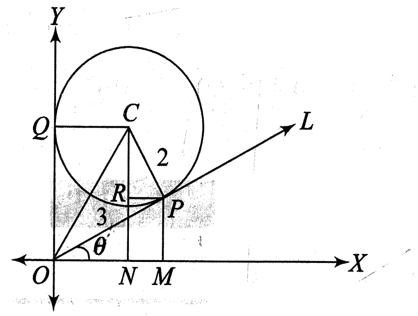
$(x - 7)^2 + (y - 6)^2 = 3^2$

& $(x - 4) + (y - 3)^2 = 3^2$

$\Rightarrow \quad x^2 + y^2 - 8x - 6y + 16 = 0$

and $\quad x^2 + y^2 - 14x - 12y + 76 = 0$

9.



Let $\angle PQX = \theta$,
then $\angle NCP = \theta$

Here, $CP = 2$, $OC = \sqrt{2^2 + 3^2} = \sqrt{13}$

$OP = \sqrt{OC^2 - OP^2} = \sqrt{13 - 4} = \sqrt{9} = 3$

$C = (2, 3)$ and $ON = 2$

Now, $OM = ON + MN = ON + PR$

$\Rightarrow \quad OP \cos\theta = 2 + 2 \sin\theta$

$\Rightarrow \quad 3 \cos\theta = 2 + 2 \sin\theta$

$\Rightarrow \quad 9\ \text{ocs}^2\theta = 4 + 4\sin^2\theta + 8 \sin\theta$

$\Rightarrow \quad 9 - 9\sin^2\theta = 4 + 4\sin^2\theta + 8\sin\theta$

$\Rightarrow \quad 13\sin^2\theta + 8\sin\theta - 5 = 0$

$\Rightarrow \quad 13\sin^2\theta + 13\sin\theta - 5\sin\theta - 5 = \theta$

$\Rightarrow \quad 13\sin\theta(\sin\theta + 1) - 5(\sin\theta + 1) = 0$

$\Rightarrow \quad (\sin\theta + 1)(13\sin\theta - 5) = 0$

$\Rightarrow \quad \sin\theta = \dfrac{5}{13}$

$\Rightarrow \quad \cos\theta = \dfrac{12}{13}$

Therefore,

$$P = (OP\cos\theta, OP\sin\theta) = \left(\frac{36}{13}, \frac{15}{13}\right)$$

10. Hence, the equation of the circle is
$$(x-2)(x-6)+(y-3)(y-9)=0$$
$$\Rightarrow\quad x^2+y^2-8x-12y+39=0$$

11. Hence, the equation of the circle be
$$x^2+y^2+2gx+2fy+c=0 \quad \text{...(i)}$$
which is passing through (1, 2), (4, 5) and (0, 9).
Thus when $(x, y) = (1, 21)$ $1+4+2g+8f+c=0$
$$\Rightarrow\quad 2g+8f+c=-5 \quad \text{...(ii)}$$
when $(x, y) = (4, 5)$
$$16+25+8g+10f+c=0$$
$$\Rightarrow\quad 8g+10f+c=-41 \quad \text{...(iii)}$$
when $(x, y) = (0, 9)$
$$0+81+18f+c=0$$
$$\Rightarrow\quad 18f+c=-81 \quad \text{...(iv)}$$
Eqs (iii) – Eqs (ii), we get
$$6g+2f=-36$$
$$\Rightarrow\quad 3g+f=-18 \quad \text{...(v)}$$
Eqs (iii) – Eqs (iv), we get
$$8g-8f=40$$
$$\Rightarrow\quad g-f=5 \quad \text{...(vi)}$$
From Eqs (v) and (vi), we get
$$4g=-13$$
$$\Rightarrow\quad g=-13/4$$
and $f=g-5=-13/4-5$
$$\Rightarrow\quad f=-33/4$$
Now, from Eq. (iv), we get,
$$c=-81-18f$$
$$=-81-297/2=-135/2$$
Put all these values of f, g and c in Eq. (i), we get
$$x^2+y^2-\frac{13}{2}x-\frac{33}{2}y-\frac{135}{2}=0$$
$$\Rightarrow\quad 2(x^2+y^2)-13x-33y-135=0$$

12. Let the equation of the circle is
$$x^2+y^2+2gx+2fy+c=0$$
which is passing through (1, 2) and (3, 4).
Now, when $(x, y) = (1, 2)$ $1+4+2g+4f+c=0$
$$\Rightarrow\quad 2g+4f+c=-5 \quad \text{(i)}$$
And, when $(x, y) = (3, 4)$ $9+16+6g+8f+c=0$
$$\Rightarrow\quad 6g+8f+c=-25 \quad \text{(ii)}$$
Eq. (ii) – Eq. (i), we get
$$4g+4f=-20$$
$$\Rightarrow\quad g+f=-5 \quad \text{(ii)}$$
Also, $$\left|\frac{-3g-f-5}{\sqrt{3^2+1^2}}\right|=\sqrt{g^2+f^2-c}$$
$$=\sqrt{(g+1)^2+(f+2)^2}$$
$$\Rightarrow\quad (3g+f+5)^{20}=10\{(g+1)^2+(g+3)^2\}$$
$$\Rightarrow\quad (2g-5+5)^2=10(g^2+2g+1+g^2+6g+9))$$
$$\Rightarrow\quad 4g^2=10(2g^2+8g+10)$$
$$\Rightarrow\quad 4g^2+20g+25=0$$
$$\Rightarrow\quad (2g+5)^2=0$$
$$\Rightarrow\quad g=-5/2$$
Thus, $f=-5-g=-5+\frac{5}{2}=-\frac{5}{2}$ and $c=10$
Hence, the equation of the required circle is
$$x^2+y^2-2\left(\frac{5}{2}\right)x-2\left(\frac{5}{2}\right)+10=0$$
$$\Rightarrow\quad x^2+y^2-5x-5y+10=0$$

13. The lengths of the x-intercept
$$=2\sqrt{g^2-c}=2\sqrt{9-8}=2$$
and the y-intercept
$$=2\sqrt{f^2-c}=2\sqrt{25-8}=2\sqrt{17}$$

14. The length of the y-intercept
$$=2\sqrt{f^2-c}=2\sqrt{\frac{1}{4}-0}=2\times\frac{1}{2}=1$$

15. The given equation of a circle is
$$x^2+y^2-2ax-2ay+a^2=0$$
$$\Rightarrow\quad (x-a)^2+(y-a)^2=a^2.$$
Thus, the centre is (a, a) and the radius is a.

16. Let $N=x^2+y^2-4x+2y-11$
The values of N at (1, 2) is
$$1+4-4+4-11=5-11=-6<0$$
and at (6, 0) is
$$36+0-0+12-11=37>0$$
Thus, the point (1, 2) lies inside of the circle and (6, 0) lies outside of the circle.

17. Since, the point $(\lambda, -\lambda)$ lies inside the circle $x^2+y^2-4x+2y-8=0$, so
$$\lambda^2+\lambda^2-4\lambda-2\lambda-8<0$$
$$\Rightarrow\quad 2\lambda^2-6\lambda-8<0$$
$$\Rightarrow\quad \lambda^2-3\lambda-4<0$$
$$\Rightarrow\quad (\lambda-4)(\lambda+1)<0$$
$$\Rightarrow\quad -1<\lambda<4$$
Thus, the range of λ is $(-1, 4)$.

18. Let $N=x^2+y^2-14x-10y-151$
The value of N at $(2, -7)$ is
$$4+49-28+98-151=153-151-28$$
$$=2-28=-26.$$
Thus, the point $P(2, -7)$ lies inside of the circle.
The centre of the circle is $C(7, 5)$ and the radius is
$$=CA=CB=15.$$
Now,
$$CP=\sqrt{(7-2)^2+(5+7)^2}$$
$$=\sqrt{25+144}=\sqrt{169}=13$$
Hence, the shortest distance,
$$PA=CA-CP=15-13=2$$
and the longest distance
$$=PB=CP+CB=13+15=28.$$

19. The line $y=x+2$ touches the circle $x^2+y^2=2$, if the length of the perpendicular from the centre to the given line is equal to the radius of a circle.

Thus, the radius of the circle is $\sqrt{2}$.

The length of the perpendicular from the centre (0, 0) to the line $x - y + 2 = 0$ is

$$\left|\frac{0-0+2}{\sqrt{1^2+1^2}}\right| = \frac{2}{\sqrt{2}} = \sqrt{2} = \text{radius.}$$

Hence, the line $y = x + 2$ touches the circle $x^2 + y^2 = 2$.

On solving $y = x + 2$ and $x^2 + y^2 = 2$, we get $x^2 + (x + 2)^2 = 2$

$\Rightarrow \quad x^2 + x^2 + 4x + 4 - 2 = 0$

$\Rightarrow \quad 2x^2 + 4x + 2 = 0$

$\Rightarrow \quad x^2 + 2x + 1 = 0$

$\Rightarrow \quad (x + 1)^2 = 0$

$\Rightarrow \quad x = -1, y = 1 + 2 = 1$

Thus, the co-ordinates of the point of contact is (–1, 1).

20. The equation of any line parallel to the line

$3x + 2y + 5 = 0$ is $3x + 2y + k = 0$ ...(i)

The line (i) will be a tangent to the circle $x^2 + y^2 = 4$, if the length of the perpendicular from the centre to the line (i) is equal to the radius of the circle.

Therefore, $\left|\frac{3.0+2.0+k}{\sqrt{3^2+2^2}}\right| = 2$

$\Rightarrow \quad k = \pm 2\sqrt{13}$

Hence, the equation of the tangents to the given circle are $3x + 2y \pm 2\sqrt{13} = 0$.

21. The equation of any line perpendicular to the line

$4x + 3y = 0$ is $3x - 4y + k = 0$ ...(i)

The line (i) will be a tangent to the circle $x^2 + y^2 = 9$, if the length of the perpendicular from the centre to the line (i) is equal to the radius of the circle.

Therefore, $\left|\frac{3.0-4.0+k}{\sqrt{3^2+4^2}}\right| = 3$

$\Rightarrow \quad k = \pm 5$

Hence, the equation of the tangents are

$3x - 4y + 15 = 0$ and $3x - 4y - 15 = 0$

22. Let the equation of line be

$y = mx + c = \sqrt{3}x + c$ ...(i)

$(\because m = \tan(60°) = \sqrt{3})$

The line (i) will be a tangent to the circle $x^2 + y^2 + 4x + 3 = 0$, if the length of the perpendicular from the centre (–2, 0) to the line (i) is equal to the radius (= 1) of a circle.

Therefore, $\left|\frac{-2\sqrt{3}-0+c}{\sqrt{3+1}}\right| = 1$

$\Rightarrow \quad c - 2\sqrt{3} = \pm 2$

$\Rightarrow \quad c = 2\sqrt{3} \pm 2$

Hence, the equation of the tangents becomes

$y = \sqrt{3}x + (2\sqrt{3} \pm 2)$

23. A line is equally inclined to the co-ordinate axes is

$y = \pm x + c$ ...(i)

The line (i) will be tangent to the circle $x^2 + y^2 = 4$, if the length of the perpendicular from the centre (0, 0) to the circle is equal to the radius (= 2) of the given circle.

Therefore, $\left|\frac{0+0+c}{\sqrt{1^2+1^1}}\right| = 2$

$\Rightarrow \quad c = \pm 4$

Hence, the equation of the tangents becomes

$y = \pm x \pm 4$

24. The equation of any line passing through (7, 1) is

$y - 1 = m(x - 7)$ ...(i)

The line (i) will be a tangent to the circle $x^2 + y^2 = 25$, if the length of the perpendicular from the centre of the given circle is equal to the radius of the same circle.

Therefore, $\left|\frac{0-0-(7m-1)}{\sqrt{m^2+1}}\right| = 5$

$\Rightarrow \quad (7m - 1)^2 = 25(m^2 + 1)$

$\Rightarrow \quad 12m^2 - 7m - 12 = 0$

$\Rightarrow \quad 12m^2 - 16m + 9m - 12 = 0$

$\Rightarrow \quad (3m - 4)(4m + 3) = 0$

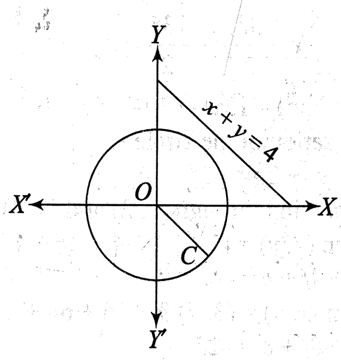
$\Rightarrow \quad m = \frac{4}{3}, -\frac{3}{4}$

Hence, the equation of the tangents are

$$y - 1 = \frac{4}{3}(x - 7) \text{ and } y - 1 = -\frac{3}{4}(x - 7)$$

$\Rightarrow \quad 4x - 3y = 25$ and $3x + 4y = 25$

25. The centre of a circle is (0, 0) and the radius is 3. Here $OC = 2$.



We shall find the co-ordinates of C.

Let C be (x, y).

Then $x = 2\cos(-45°) = \sqrt{2}$

and $y = 2\sin(-45°) = -\sqrt{2}$.

Thus, the co-ordinates of C becomes $(\sqrt{2}, -\sqrt{2})$.

Hence, the equation of a circle is

$(x - \sqrt{2})^2 + (y + \sqrt{2})^2 = 3^2$

$\Rightarrow \quad x^2 + y^2 - 2\sqrt{2}x - 2\sqrt{2}y - 5 = 0$

26. When $x = 2$, then $y = \pm\sqrt{5}$.

Therefore, the points are $(2, \sqrt{5})$ and $(2, -\sqrt{5})$.

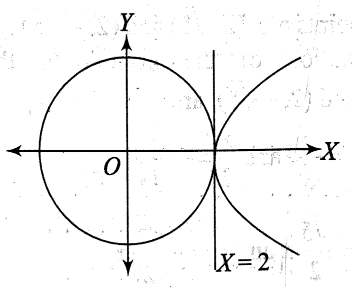
Thus, the equation of the tangents at $(2, \sqrt{5})$ and $(2, -\sqrt{5})$ are $2x + \sqrt{5}y = 9$ and $2x - \sqrt{5}y = 9$.

27. When $y = 4$, then $x = 0$.
Therefore, the point is $(0, 4)$.
Hence, the equation of the tangent to the given circle at $(0, 4)$ is $x \cdot 0 + y \cdot 4 = 16$
$\Rightarrow \quad y = 4.$

28. The given equations are
$$4x - 3y = 10 \qquad \ldots(i)$$
and $\quad x^2 + y^2 - 2x + 4y - 20 = 0 \qquad \ldots(ii)$
On solving Eqs (i) and (ii), we get
$$x^2 + \left(\frac{4x-3}{10}\right)^2 - 2x + 4\left(\frac{4x-3}{10}\right) - 20 = 0$$
$\Rightarrow \quad 9x^2 + (4x - 10)^2 - 18x + 48x - 120 - 180 = 0$
$\Rightarrow \quad 25x^2 - 50x - 200 = 0$
$\Rightarrow \quad x^2 - 2x - 8 = 0$
$\Rightarrow \quad (x - 4)(x + 2) = 0$
$\Rightarrow \quad x = -2, 4$ and $y = -\frac{11}{10}, \frac{13}{10}$
Hence, the points of intersection are
$$\left(-2, -\frac{11}{10}\right) \text{ and } \left(4, \frac{13}{10}\right)$$

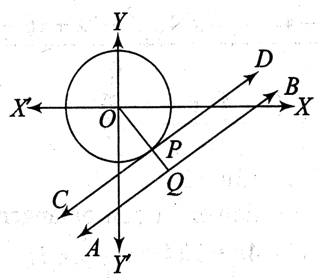
29. The given circle is $x^2 + y^2 - 2x + 4y = 0$
$\Rightarrow \quad (x - 1)^2 + (y + 2)^2 = 5 \qquad \ldots(i)$
The equation of any line passing through $(0, 1)$ is
$$y - 1 = m(x - 0) \qquad \ldots(ii)$$
The line (ii) will be a tangent to the circle (i), then the length of the perpendicular from the centre $(1, -2)$ to the line (ii) is equal to the radius of a circle (i).
Therefore, $\left|\frac{m + 2 + 1}{\sqrt{m^2 + 1}}\right| = \sqrt{5}$
$\Rightarrow \quad (m + 3)^2 = 5(m^2 + 1)$
$\Rightarrow \quad m^2 + 5m + 9 - 5m^2 - 5 = 0$
$\Rightarrow \quad 2m^2 - 3m - 2 = 0$
$\Rightarrow \quad (m - 2)(2m + 1) = 0$
$\Rightarrow \quad m = -\frac{1}{2}, 2$
Hence, the equation of the tangents are
$$y - 1 = -\frac{1}{2}x \text{ and } y - 1 = 2x$$
$\Rightarrow \quad x + 2y - 1 = 0$ and $2x - y + 1 = 0$

30. The given curves are $x^2 + y^2 = 4$ and $y^2 = 4(x - 2)$.



From the graph, it is clear that the equation of the common tangent is $x = 2$.

31. $x^2 + y^2 - 2x + 4y - 20 = 0$



Let AB: $y = x - 4$ and CD be a tangent parallel to AB.
Now $OQ = \left|\frac{0 - 0 - 8}{\sqrt{1^2 + 1^2}}\right| = \frac{8}{\sqrt{2}} = 4\sqrt{2}$
and OP = radius of a circle = 3.
Thus, the shortest distance $= PQ$
$= OQ - OP$
$= (4\sqrt{2} - 3)$

32. Hence, the length of the tangent from the point $(2, 3)$ to the circle $x^2 + y^2 = 4$ is
$$\sqrt{4 + 9 - 4} = \sqrt{9} = 3$$

33. Let $P(x_1, y_1)$ be any point on the circle $x^2 + y^2 = a^2$.
Therefore, $x_1^2 + y_1^2 = a^2$.
Now, the length of the tangent from (x_1, y_1) to the circle $x^2 + y^2 = b^2$ is
$$\sqrt{x_1^2 + y_1^2 - b^2} = \sqrt{a^2 - b^2}.$$

34. Let $P(x_1, y_1)$ be any point on
$$x^2 + y^2 + 2011x + 2012y + 2013 = 0$$
Therefore,
$$x_1^2 + y_1^2 + 2011x_1 + 2012y_1 + 2013 = 0$$
Now, the length of the tangent from (x_1, y_1) to the circle $x^2 + y^2 + 2011x + 2012y + 2014 = 0$ is
$$\sqrt{x_1^2 + y_1^2 + 2011x_1 + 2012y_1 + 2014}$$
$$= \sqrt{-2013 + 2014}$$
$$= 1$$

35. The power of a point $(2, 5)$ with respect to a circle $x^2 + y^2 = 16$ is $(4 + 25 - 16) = 13$.

36. Let the point P represents a complex number Z. i.e. $Z = 1 + 2i$
Then, from the rotation theorem,
$$Q = iZ = i(1 + 2i) = i - 2 = -2 + 1.i = (-2, 1)$$
Thus, the power of a point Q with respect to a circle is $4 + 1 - 4 = 1$.

37. Hence, the equation of a pair of tangents to a circle $x^2 + y^2 = 4$ from a point $(1, 2)$ is $SS_1 = T^2$
$\Rightarrow \quad (x^2 + y^2 - 4)(1 + 4 - 4) = (x + 2y - 4)^2$
$\Rightarrow \quad (x^2 + y^2 - 4) = (x^2 + 4y^2 + 16 + 4xy - 8x - 16y)$

$\Rightarrow \quad 3y^2 + 4(x-4)y + 4(5-2x) = 0$

$$\Rightarrow \quad y = \frac{-4(x-4) \pm \sqrt{16(x-4)^2 - 48(5-2x)}}{6}$$

$$\Rightarrow \quad y = \frac{-4(x-4) \pm 4\sqrt{x^2 - 8x + 16 - 15 + 6x}}{6}$$

$$\Rightarrow \quad y = \frac{-4(x-4) \pm 4(x-1)}{6}$$

$\Rightarrow \quad y = 2, 4x - 3y + 2 = 0$

38. Hence, the equation of the pair of tangents are $SS_1 = T^2$

$\Rightarrow \quad (x^2 + y^2 - 4x + 3)(4 + 9 - 8 + 3)$

$= (2x + 3y - 2(x+2) + 3)^2$

$\Rightarrow \quad 8(x^2 + y^2 - 4x + 3) = (3y - 1)^2$

$\Rightarrow \quad 8(x^2 + y^2 - 4x + 3) = 9y^2 - 6y + 1$

$\Rightarrow \quad y^2 - 8x^2 + 32x - 6y - 23 = 0$

$\Rightarrow \quad y^2 - 6y - (8x^2 - 32x + 23) = 0$

$$\Rightarrow \quad y = \frac{6 \pm \sqrt{36 + 4(8x^2 - 32x + 23)}}{2}$$

$$\Rightarrow \quad y = \frac{6 \pm \sqrt{32x^2 - 128x + 128}}{2}$$

$$= \frac{6 \pm \sqrt{32(x^2 - 4x + 4)}}{2}$$

$$= \frac{6 \pm 4\sqrt{2}(x-2)}{2}$$

$\Rightarrow \quad y = 3 \pm 2\sqrt{2}(x-2)$

39. Hence, the angle between the tangent from the point (3, 5) to the circle $x^2 + y^2 = 25$ is

$$2\tan^{-1}\left(\frac{a}{\sqrt{S_1}}\right) = 2\tan^{-1}\left(\frac{5}{\sqrt{9+25-25}}\right)$$

$$= 2\tan^{-1}\left(\frac{5}{3}\right)$$

40. After translation the point (1, 2) becomes

$(1+2, 2) = (3, 2)$.

Hence, the angle between the tangent from the point (3, 2) to the circle $x^2 + y^2 = 9$ is

$$2\tan^{-1}\left(\frac{a}{\sqrt{S_1}}\right) = 2\tan^{-1}\left(\frac{3}{\sqrt{9+4-9}}\right)$$

$$= 2\tan^{-1}\left(\frac{3}{2}\right)$$

41. The locus of the point of intersection of two perpendicular tangents to a circle is the director circle of the given circle.

Thus, the equation of the director circle to the circle $x^2 + y^2 = 25$ is $x^2 + y^2 = 2 \times 25 = 50$

42. Clearly, the locus of the arbitrary point is the director circle of the given circle $x^2 + y^2 = 9$.

Hence, the equation of the director circle to the circle $x^2 + y^2 = 9$ is $x^2 + y^2 = 2 \times 9 = 18$.

43. Clearly, the equation $x^2 + y^2 = 20$ is the director circle of the circle $x^2 + y^2 = 10$.

Hence, the angle between the pair of tangents is 90°.

44. (i) The given circle $x^2 + y^2 + 2x = 0$ can be reduced to $(x+1)^2 + y^2 = 1$.

Hence, the equation of the director circle to the circle $(x+1)^2 + y^2 = 1$ is

$(x+1)^2 + y^2 = 2 \times 1 = 2$

$\Rightarrow \quad x^2 + y^2 + 2x - 1 = 0$

(ii) The given circle $x^2 + y^2 + 10y + 24 = 0$ can be reduced to $x^2 + (y+5)^2 = 1$

Hence, the equation of the director circle to the circle $x^2 + (y+5)^2 = 1$ is

$x^2 + (y+5)^2 = 2 \times 1 = 1$

$\Rightarrow x^2 + y^2 + 10y + 23 = 0$

(iii) The given circle

$x^2 + y^2 + 16x + 12y + 99 = 0$

can be reduced to

$(x+8)^2 + (y+6)^2 = 64 + 36 - 99 = 1$

Hence, the equation of the director circle to the circle $(x+8)^2 + (y+6)^2 = 1$

is $(x+8)^2 + (y+6)^2 = 2 \times 1 = 2$

$\Rightarrow \quad x^2 + y^2 + 16x + 12y + 98 = 0$

(iv) The given circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ can be reduced to

$(x+g)^2 + (y+f)^2 = (\sqrt{g^2 + f^2 - c})^2$

Hence, the equation of the director circle to the circle $(x+g)^2 + (y+f)^2 = (\sqrt{g^2 + f^2 - c})^2$ is

$(x+g)^2 + (y+f)^2 = 2(g^2 + f^2 - c)$.

$\Rightarrow \quad x^2 + y^2 + 2gx + 2fy - g^2 - f^2 + 2c = 0$.

(v) The given circle $x^2 + y^2 - ax - by = 0$ can be reduced to $\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \left(\frac{\sqrt{a^2+b^2}}{2}\right)^2$

Hence, the equation of the director circle to the circle

$$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \left(\frac{\sqrt{a^2+b^2}}{2}\right)^2 \text{ is}$$

$$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \left(\frac{a^2+b^2}{2}\right)$$

$$\Rightarrow x^2 + y^2 - ax - by - \frac{1}{4}(a^2 + b^2) = 0$$

45. When $x = 2, y = \pm\sqrt{5}$.

So, the points are $(2, \sqrt{5})$ and $(2, -\sqrt{5})$.

The equation of the normal to the circle at $(2, \sqrt{5})$ and $(2, -\sqrt{5})$ are

$$\frac{x}{2} = \frac{y}{\sqrt{5}} \text{ and } \frac{x}{2} = -\frac{y}{\sqrt{5}}$$

$$\Rightarrow \quad y = \frac{\sqrt{5}}{2}x \text{ and } y = -\frac{\sqrt{5}}{2}x$$

46. The equation of the normal to the given circle at (–2, 1) is

$$\Rightarrow \quad \frac{x-x_1}{x_1+g}=\frac{y-y_1}{y_1+f}$$

$$\frac{x+2}{-2+1}=\frac{y-1}{1+2}$$

$\Rightarrow \quad 3x+6=-y+1$

$\Rightarrow \quad 3x+y+5=0$

47. The centre of a circle $x^2+y^2-4x-6y+4=0$ is (2, 3).
The equation of a normal parallel to $x-y-3=0$ is

$x-y+k=0 \quad$ ...(i)

As we know that the normal always passes through the centre of a circle.
Therefore, $2-3+k=0 \Rightarrow k=1$.
Hence, the equation of the normal is

$x-y+1=0$

48. The centre of a circle

$x^2+y^2-8x-12y+99=0$ is (4, 6).

The equation of the normal which is perpendicular to

$2x-3y+10=0$ is $3x+2y-k=0 \quad$ ...(i)

which is passing through (4, 6).
Therefore, $12+12-k=0 \Rightarrow k=24$.
Hence, the equation of the normal is $3x+2y=24$.

49. The equation of the chord of contact of tangents from (5, 3) to the circle $x^2+y^2=25$ is $5x+3y=25$.

50. Let the point be (x_1, y_1).
Clearly, $2x+y+12=0 \quad$ ...(i)
is the chord of contact of the tangents to the circle $x^2+y^2-4x+3y-1=0$ from (x_1, y_1).
The equation of the chord of contact of tangents to the circle $x^2+y^2-4x+3y-1=0$ from the point (x_1, y_1) is

$$xx_1+yy_1-2(x+x_1)+\frac{3}{2}(y+y_1)-1=0 \quad \text{...(ii)}$$

Clearly Eqs (i) and (ii) are identical.

$$\text{Therefore, } \frac{x_1-2}{2}=\frac{y_1+\frac{3}{2}}{1}=\frac{\frac{3}{2}y_1-2x_1-1}{12}$$

$\Rightarrow \quad x_1-2y_1=5$ and $4x_1+21y_1=-38$

$\Rightarrow \quad x_1=1, y_1=-2$

Hence, the point is (1, –2).

51. The equation of the chord of contact QR is

$hx+ky=a^2 \quad$ ...(i)

The chord of contact subtends a right angle at the centre of the circle $x^2+y^2=a^2$.

$$\text{Therefore, } \frac{x^2+y^2}{a^2}=\left(\frac{hx+ky}{a^2}\right)^2$$

$\Rightarrow \quad a^2(x^2+y^2)=(hx+ky)^2$

$\Rightarrow \quad x^2(a^2-h^2)+y^2(a^2-k^2)-2hkxy=0$

Since, the line (i) makes a right angle at the centre, so $(a^2-h^2)+(a^2-k^2)=0$

$\Rightarrow \quad h^2+k^2=2a^2$

which is the required condition.

52. Let AB be the chord of contact.
Then $AB: hx+ky=a^2$

$$\text{Now, } PM=\left|\frac{h^2+k^2-a^2}{\sqrt{h^2+k^2}}\right|$$

The length of the chord of contact,

$$AB=\frac{2LR}{\sqrt{L^2+R^2}},$$

where L = the length of the tangent and R be the radius of the circle.
Therefore,

$$AB=\frac{2a\times\sqrt{h^2+k^2-a^2}}{\sqrt{(h^2+k^2-a^2)+a^2}}$$

$$=\frac{2a\times\sqrt{h^2+k^2-a^2}}{\sqrt{h^2+k^2}}$$

Thus, the area of the $\Delta\, PAB$

$$=\frac{1}{2}\times\left(\frac{2a\times\sqrt{h^2+k^2-a^2}}{\sqrt{h^2+k^2}}\right)\times\left(\frac{h^2+k^2-a^2}{\sqrt{h^2+k^2}}\right)$$

$$=a\left(\frac{(h^2+k^2-a^2)^{3/2}}{(h^2+k^2)}\right)$$

53. Let $P(a\cos\theta, a\sin\theta)$ be any point on the circle $x^2+y^2=a^2$.
Therefore, QR:

$ax\cos\theta+ay\sin\theta=b^2 \quad$ ...(i)

Since the line (i) be a tangent to the circle $x^2+y^2=c^2$, so

$$\left|\frac{0+0-b^2}{\sqrt{a^2\cos^2\theta+a^2\sin^2\theta}}\right|=c$$

$\Rightarrow \quad b^2=ac$

Thus, $m=2$, $n=1$ and $p=1$.
Hence, the value of $m+n+p+10=14$.

54. The equation of the chord of the circle $x^2+y^2=25$ bisected at (–2, 3) is

$T=S_1$

$\Rightarrow \quad -2x+3y-25=4+9-25$

$\Rightarrow \quad -2x+3y=13$

$\Rightarrow \quad 2x-3y+13=0$

55. Hence, the equation of the chord of the circle $x^2+y^2+6x+8y-11=0$, whose mid-point is (1, –1) is $T=S_1$

$\Rightarrow \quad x-y+3(x+1)+4(y-1)-11 = 1+1+6+8-11$

$\Rightarrow \quad 4x+3y-1=16$

$\Rightarrow \quad 4x+3y=17$

56. Let the chord AB is bisected at $C(x_1, y_1)$

The equation of AB is $xx_1 + yy_1 = x_1^2 + y_1^2$

which is passing through (h, k), so

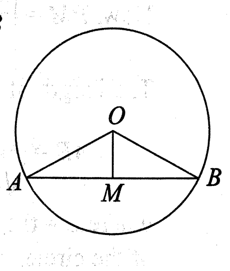
$$hx_1 + ky_1 = x_1^2 + y_1^2$$

Thus, the locus of (x_1, y_1) is

$$x^2 + y^2 = hx + ky$$

57. Let $M(h, k)$ is the mid-point of AB

In ΔOAM,



$$\sin(45°) = \frac{OM}{AM} = \frac{OM}{2}$$

$$\Rightarrow \quad OM = 2\sin(45°)$$

$$= 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\Rightarrow \quad OM^2 = 2$$

$$\Rightarrow \quad h^2 + k^2 = 2$$

Hence, the locus of (h, k) is $x^2 + y^2 = 2$.

58. Let A represents a complex number Z.

i.e. $Z = \sqrt{3} + i$

Here, $\angle AOB = 90°$.

By rotation theorem, we can say that

$$B = iZ = i(\sqrt{3} + i) = -1 + i\sqrt{3}$$

Hence, the point B is $(-1, \sqrt{3})$.

59. Let C represents a complex number Z and D be Z_1.

Here, $\angle COD = 60°$

By rotation theorem, $\dfrac{Z_1 - 0}{Z - 0} = \left|\dfrac{Z_1 - 0}{Z - 0}\right| \times e^{-i\frac{\pi}{3}}$

$$\Rightarrow \quad \frac{Z_1}{Z} = \frac{|Z_1|}{|Z|} \times e^{-i\frac{\pi}{3}} = e^{-i\frac{\pi}{3}}$$

$$\Rightarrow \quad Z_1 = Ze^{-i\frac{\pi}{3}}$$

$$\Rightarrow \quad Z_1 = (2\sqrt{2} + i)\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow \quad Z_1 = \left(\sqrt{2} + \frac{\sqrt{3}}{2}\right) + i\left(\frac{1}{2} - \sqrt{3}\right)$$

Hence, the co-ordinates of D are

$$\left(\sqrt{2} + \frac{\sqrt{3}}{2}, \frac{1}{2} - \sqrt{3}\right)$$

60. As we know that the locus of the mid-points of the chord of the circle is the diameter of the circle. Thus, the equation of the diameter which is parallel to

$$2012x + 2013y + 2014 = 0$$

is $\quad 2012x + 2013y + k = 0$

which is passing through the centre of the circle $x^2 + y^2 = 9$.

Therefore, $k = 0$

Hence, the equation of the diameter is

$$2012x + 2013y = 0$$

61. Obviously, the locus of the mid-point of the chords of the circle is the diameter of the given circle.

Here, the centre of the circle $x^2 + y^2 - 4x - 6y = 0$ is $(2, 3)$.

The equation of the diameter of the circle, which is perpendicular to the line $4x + 5y + 10 = 0$ is $5x - 4y + k = 0$ which is passing through $(2, 3)$.

Therefore, $k = 12 - 10 = 2$.

Hence, the equation of the diameter is

$$5x - 4y + 2 = 0$$

62. The equation of the common chord of the circle is

$$2x - 2y = 0$$

$$\Rightarrow \quad x = y$$

On solving $y = x$ and the circle

$$x^2 + y^2 + 3x + 5y + 4 = 0,$$

we get the points of intersection.

Thus, $x = -2 + \sqrt{2} = y$ and $x = -2 - \sqrt{2} = y$

Let the common chord be PQ, where

$$P = (-2 + \sqrt{2}, -2 + \sqrt{2}) \text{ and}$$

$$Q = (-2 - \sqrt{2}, -2 - \sqrt{2})$$

Thus, the length of the common chord, PQ

$$= \sqrt{(-2 - \sqrt{2} + 2 - \sqrt{2})^2 + (-2 - \sqrt{2} + 2 - \sqrt{2})^2}$$

$$= \sqrt{8 + 8} = \sqrt{16} = 4$$

63. The equation of the common chord of the given circles is $2x + 1 = 0$.

On solving $2x + 1 = 0$ and

$$x^2 + y^2 + 2x + 3y + 1 = 0,$$

we get the points of intersection.

Thus, the points of intersection are

$$x = -\frac{1}{2} \text{ and } y = -\frac{3}{2} \pm \sqrt{2}$$

Let PQ be the common chord, where

$$P = \left(-\frac{1}{2}, -\frac{3}{2} + \sqrt{2}\right) \text{ and}$$

$$Q = \left(-\frac{1}{2}, -\frac{3}{2} - \sqrt{2}\right)$$

The mid-point of P and Q is the centre of the new circle.

Thus, centre is $C = \left(-\frac{1}{2}, -\frac{3}{2}\right)$

Now, radius $r = CP = \sqrt{2}$.

Hence, the equation of the circle is

$$\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{3}{2}\right)^2 = 2$$

$$\Rightarrow \quad 2(x^2 + y^2) + 2x + 6y + 1 = 0$$

64. (i) The given circles are $x^2 + y^2 = 4$ ...(i)

and $x^2 + y^2 - 2x = 0$ (ii)

Now, $C_1(0, 0)$ and $C_2(1, 0)$ and $r_1 = 2$ and $r_2 = 1$.

Thus, $C_1C_2 = 1 = r_2 - r_1$

Hence, both the circles touch each other internally.

Thus, the number of common tangent = 1.

(ii) The given circles are $x^2 + y^2 + 4x + 6y + 12 = 0$

and $x^2 + y^2 - 6x - 4y + 12 = 0$

Now, C_1: (–2, –3), C_2: (3, 2), $r_1 = 1$ and $r_2 = 1$

Therefore,

$$C_1C_2 = \sqrt{(3+2)^2 + (-3-2)^2}$$
$$= \sqrt{25+25} = \sqrt{50} = 5\sqrt{2}$$

and $r_1 + r_2 = 1 + 1 = 2$.

Thus, $C_1C_2 > r_1 + r_2$

Hence, the number of common tangents = 4.

(iii) The given circles are

$x^2 + y^2 - 6x - 6y + 9 = 0$

and $x^2 + y^2 + 6x + 6y + 9 = 0$

Now, $C_1 = (3, 3)$, $C_2 = (-3, -3)$, $r_1 = 3$ and $r_2 = 3$.

Therefore,

$$C_1C_2 = \sqrt{(-3-2)^2 + (-3-3)^2}$$
$$= \sqrt{36+36} = \sqrt{72} = 6\sqrt{2}$$

and $r_1 + r_2 = 3 + 3 = 6$

Thus, $C_1C_2 > r_1 + r_2$

Hence, the number of common tangents = 4.

(iv) The given circles are

$x^2 + y^2 - 4x - 4y = 0$

and $x^2 + y^2 + 2x + 2y = 0$

Now, $C_1 = (2, 2)$, $C_2 = (-1, -1)$, $r_1 = 2\sqrt{2}$ and $r_2 = \sqrt{2}$

Therefore,

$$C_1C_2 = \sqrt{(-1-2)^2 + (-1-2)^2}$$
$$= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

and $r_1 + r_2 = 2\sqrt{2} + \sqrt{2} = 3\sqrt{2}$

Thus, $C_1C_2 = r_1 + r_2$

Hence, the number of common tangents = 3.

(v) The given circles are $x^2 + y^2 = 64$

and $x^2 + y^2 - 4x - 4y + 4 = 0$

Now, C_1: (0, 0), C_2: (2, 2), $r_1 = 8$ and $r_2 = 2$

Therefore,

$$C_1C_2 = \sqrt{(2-0)^2 + (2-0)^2}$$
$$= \sqrt{4+4} = 2\sqrt{2}$$

and $r_1 + r_2 = 8 + 2 = 10$

Therefore, $C_1C_2 < r_1 + r_2$

Thus, one circle lies inside of the other.

Hence, the number of common tangents = 0.

(vi) The given circles are

$$x^2 + y^2 - 2(1+\sqrt{2})x - 2y + 1 = 0$$

and $x^2 + y^2 - 2x - 2y + 1 = 0$

Now C_1: $(1+\sqrt{2}, 1)$, C_2: (1, 1),

$r_1 = \sqrt{2} + 1$ and $r_2 = 1$.

Therefore, $C_1C_2 = \sqrt{(\sqrt{2})^2} = \sqrt{2}$

and $r_1 - r_2 = \sqrt{2}$.

Thus, $C_1C_2 = r_1 - r_2$

Hence, the number of common tangents = 1.

65. The given circles are

$x^2 + y^2 - 2x - 6y + 9 = 0$

and $x^2 + y^2 + 6x - 2y + 1 = 0$

Now, C_1: (1, 3), C_2: (–3, 1), $r_1 = 1$ and $r_2 = 3$.

Therefore,

$$C_1C_2 = \sqrt{(-3-1)^2 + (1-3)^2}$$
$$= \sqrt{16+4} = \sqrt{20} = 2\sqrt{5}$$

and $r_1 + r_2 = 1 + 3 = 4$

Then, $C_1C_2 > r_1 + r_2$

Thus, both the circles do not intersect.

Hence, the number of common tangents = 4, in which 2 are direct common tangents and another 2 are transverse common tangents.

Here, the point M divides $C_2(-3, 1)$ and $C_1(1, 3)$ internally in the ratio 3 : 1.

Then, the co-ordinates of M are

$$\left(\frac{3.1 - 1.3}{3+1}, \frac{3.3 + 1.1}{3+1}\right) = \left(0, \frac{5}{2}\right)$$

Also, the point P divide $C_2(-3, 1)$ and C_1 (1, 3) externally in the ratio 3 : 1.

Then, the co-ordinates of P are

$$\left(\frac{3.1 + 1.3}{3-1}, \frac{3.3 - 1.1}{3-1}\right) = (3, 4)$$

Case I: Direct common tangents

Any line through (3, 4) is

$y - 4 = m(x - 3)$ ...(i)

$\Rightarrow \quad mx - y + 4 - 3m = 0$

If it is a tangent to the circle,

$$\left|\frac{m - 3 + 4 - 3m}{\sqrt{m^2+1}}\right| = 1$$

$\Rightarrow \quad (1 - 2m)^2 = (m^2 + 1)$

$\Rightarrow \quad 3m^2 - 4m = 0$

$\Rightarrow \quad m(3m - 4) = 0$

$\Rightarrow \quad m = 0, \frac{4}{3}.$

Hence, the direct common tangents are

$y = 4$, $4x - 3y = 0$.

Case II: Transverse common tangents

Any line through (0, 5/2) is

$$y-\frac{5}{2}=m(x-0) \quad \text{...(i)}$$

$\Rightarrow \quad 2mx-2y+5=0$

If it is a tangent to a circle, then

$$\left|\frac{m\cdot 1-3+\frac{5}{2}}{\sqrt{m^2+1}}\right|=1$$

$\Rightarrow \quad (2m-1)^2=4(m^2+1)$

$\Rightarrow \quad 4m^2-4m+1=4(m^2+1)$

$\Rightarrow \quad 4m+3=0$

$\Rightarrow \quad m=\infty, m=-\frac{3}{4}$

Hence, the equation of transverse tangents are $x=0$ and $3x+4y=10$.

66. Given circles are $x^2+y^2+c^2=2ax$

$\Rightarrow \quad (x-a)^2+y^2=(\sqrt{a^2-c^2})^2 \quad$ ...(i)

and $x^2+y^2+c^2=2by$

$\Rightarrow \quad x^2+(y-b)^2=(\sqrt{b^2-c^2})^2 \quad$ ...(ii)

Now $C_1:(a, 0)$, $C_2:(0, b)$,

$r_1=\sqrt{a^2-c^2}$ and $r_2=\sqrt{b^2-c^2}$

Since both the circles touch each other externally, then $C_1C_2=r_1+r_2$

$\Rightarrow \quad \sqrt{a^2+b^2}=\sqrt{a^2-c^2}+\sqrt{b^2-c^2}$

$\Rightarrow \quad a^2+b^2=a^2-c^2+b^2-c^2+2(\sqrt{a^2-c^2})(\sqrt{b^2-c^2})$

$\Rightarrow \quad 2c^2=2(\sqrt{a^2-c^2})(\sqrt{b^2-c^2})$

$\Rightarrow \quad c^4=(a^2-c^2)(b^2-c^2)=a^2b^2-c^2(a^2+b^2)+c^4$

$\Rightarrow \quad a^2b^2-c^2(a^2+b^2)=0$

$\Rightarrow \quad \frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}$

67. The given circles are

$x^2+y^2=9 \quad$ ...(i)

and $x^2+y^2-8x-6y+n^2=0$

$\Rightarrow \quad (x-4)^2+(y-3)^2=(\sqrt{25-n^2})^2 \quad$ ...(ii)

Here, C_1: (0, 0), C_2: (4, 3), $r_1=3$ and $r_2=\sqrt{25-n^2}$

Since the circles have only two common tangents, so we can write,

$|r_1-r_2|<C_1C_2<r_1+r_2$

$\Rightarrow \quad |3-\sqrt{25-n^2}|<5<3+\sqrt{25-n^2}$

Case I: When $|3-\sqrt{25-n^2}|<5$

Then $-5<3-\sqrt{25-n^2}<5$

$\Rightarrow \quad -5-3<-\sqrt{25-n^2}<5-3$

$\Rightarrow \quad -2<\sqrt{25-n^2}<8$

$\Rightarrow \quad 4<25-n^2<64$

$\Rightarrow \quad -21<-n^2<39$

$\Rightarrow \quad -39<n^2<21$

$\Rightarrow \quad 0<n<\sqrt{21}$

$\Rightarrow \quad n=1, 2, 3, 4$

Case II: When $5<3+\sqrt{25-n^2}$

Then, $\sqrt{25-n^2}>2$

$\Rightarrow \quad 25-n^2>4$

$\Rightarrow \quad n^2<21$

$\Rightarrow \quad n<\sqrt{21}$

$\Rightarrow \quad n=1, 2, 3, 4.$

Hence, the number of integers = 4.

68. The given circles are

$x^2+y^2+x+y=0$

and $x^2+y^2+x-y=0$

Here, $g_1=-1/2, f_1=-1/2, g_2=-1/2, f_2=1/2, c_1=0, c_2=0$

Now, $2(g_1g_2+f_1f_2)=2(-1/4+1/4)=0$ and $c_1+c_2=0$.

Hence, the angle between the given circles is 90°.

69. The given circles are

$x^2+y^2+2a_1x+2b_1y+c_1=0$

and $x^2+y^2+a_2x+b_2y+c_2=0$

Here, $g_1=a_1, f_1=b_1, g_2=a_2/2, f_2=b_2/2, c_1=c_1$ and $c_2=c_2$

Since the given two circles are orthogonal, so

$2(g_1g_2+f_1f_2)=c_1+c_2$

$\Rightarrow \quad 2\left(a_1\cdot\frac{a_2}{2}+b_1\cdot\frac{b_2}{2}\right)=c_1+c_2$

$\Rightarrow \quad (a_1\cdot a_2+b_1\cdot b_2)=c_1+c_2$

Hence, the result.

70. The given circles are

$2x^2+2y^2-3x+6y+k=0$

$\Rightarrow \quad x^2+y^2-\frac{3}{2}x+3y+\frac{k}{2}=0$

and $x^2+y^2-4x+10y+16=0$

Since, the two given circles are orthogonal, then $2(g_1g_2+f_1f_2)=c_1+c_2$

$\Rightarrow \quad 2\left(-\frac{3}{4}\times -2+\frac{3}{2}\times 5\right)=\frac{k}{2}+16$

$\Rightarrow \quad \frac{k}{2}+16=18$

$\Rightarrow \quad \frac{k}{2}=18-16$

$\Rightarrow \quad k=4$

Hence, the value of k is 4.

71. Let the equation of the circle be

$x^2+y^2+2gx+2fy=0 \quad$ ...(i)

Therefore, the centre of the circle is $(-g, -f)$.

Since, the centre lies on the line $y=x$, so

$-f=-g$

$\Rightarrow \quad f=g$

The Eq. (i) is orthogonal to

$$x^2+y^2-4x-6y+18=0,$$

so, $2[g(-2)+f(-3)]=0+18$

$\Rightarrow$ $2(-5g)=18$ $(\because f=g)$

$\Rightarrow$ $g=-\dfrac{9}{5}=f$

Hence, the equation of the circle is

$$x^2+y^2-\frac{18}{5}x-\frac{18}{5}y=0$$

72. Let the equation of the circle be

$$x^2+y^2+2gx+2fy+c=0 \quad \text{...(i)}$$

The equation of the circle (i) is orthogonal to

$$x^2+y^2+4x-6y+9=0$$

and $x^2+y^2-5x+4y-2=0$

Thus, $(4g-6f)=c+9$ ...(ii)

and $(-5g+4f)=c-2$ ...(iii)

Subtracting, we get $-9g+10f+11=0$

$\Rightarrow$ $9(-g)-10(-f)+11=0$

Hence, the locus of the centre is

$$9x-10y+11=0$$

73. Let the equation of the circle be

$$x^2+y^2+2gx+2fy+c=0 \quad \text{...(i)}$$

which is passing through (a, b).

Therefore,

$$a^2+b^2+2ga+2fb+c=0 \quad \text{...(ii)}$$

It is given that the circle (i) is orthogonal to $x^2+y^2=4$,

so $2(g\cdot 0+f\cdot 0)=c-4 \Rightarrow c=4$

From Eq. (ii), we get

$$a^2+b^2+2g\cdot a+2f\cdot b+4=0$$

Hence, the locus of $(-g, -f)$ is

$$2ax+2by-(a^2+b^2+4)=0$$

74. Let C_1 and C_2 be the centres of the two orthogonal circles with radii r_1 and r_2, respectively.

Here $\angle C_1PC_2=90°$ and let

$$\angle PC_1C_2=\theta \text{ and } \angle PC_2C_1=90°-\theta$$

Thus,

$$\sin(\theta)=\frac{PM}{r_1} \quad\text{and}\quad \sin(90°-\theta)=\frac{PM}{r_2}.$$

Squaring and adding, we get

$$\Rightarrow \left(\frac{PM}{r_1}\right)^2+\left(\frac{PM}{r_2}\right)^2=1$$

$$PM^2\left(\frac{1}{r_1^2}+\frac{1}{r_2^2}\right)=1$$

$$\Rightarrow PM^2=\frac{r_1^2r_2^2}{r_1^2+r_2^2}$$

$$\Rightarrow PM=\frac{r_1r_2}{\sqrt{r_1^2+r_2^2}}$$

Hence, the length of the common chord,

$$PQ=2PM=\frac{2r_1r_2}{\sqrt{r_1^2+r_2^2}}$$

75. Hence, the equation of the radical axis is

$$(x^2+y^2+4x+6y+9)-(x^2+y^2+3x+8y+10)=0$$

$\Rightarrow$ $(4x-3x)+(6y-8y)+(9-10)=0$

$\Rightarrow$ $x-2y-1=0$

76. Hence, the equation of the radical axis is

$$(x^2+y^2+8x+2y+10)-(x^2+y^2+7x+3y+10)=0$$

$\Rightarrow$ $(8x-7x)+(2y-3y)=0$

$\Rightarrow$ $y=x$

Hence, the image of (2, 3) with respect to the line $y=x$ is (3, 2).

77. Let $S_1: x^2+y^2=1$ ...(i)

$S_2: x^2+y^2-8x+15=0$ ...(ii)

and $S_3: x^2+y^2+10y+24=0$ ...(iii)

Eq. (i) – Eq. (ii), we get

$$8x=16 \Rightarrow x=2$$

Eq. (i) – Eq. (iii), we get

$$-10y=25 \Rightarrow y=-5/2$$

Hence, the radical centre is $\left(2, -\dfrac{5}{2}\right)$.

78. Let the equation of the circle be

$$x^2+y^2+2gx+2fy+c=0 \quad \text{...(i)}$$

The circle (i) is orthogonal to

$$x^2+y^2-3x-6y+14=0,$$

$$x^2+y^2-x-4y+8=0,$$

and $x^2+y^2+2x-6y+9=0$

Therefore,

$$2\left(g\cdot\left(-\frac{3}{2}\right)-3f\right)=c+14$$

$\Rightarrow$ $(-3g-6f)=c+14$ ...(ii)

$$2\left[g\cdot\left(-\frac{1}{2}\right)-2f\right]=c+8$$

$\Rightarrow$ $(2g-4f)=c+8$ ...(iii)

and

$$2(g\cdot 1-3\cdot f)=c+9$$

$\Rightarrow$ $(2g-6f)=c+9$ ...(iv)

Eq. (iii) – Eq. (ii), we get

$$5g+2f=-6 \quad \text{...(v)}$$

Eq. (iii) – Eq. (iv), we get

$$2f=-1 \Rightarrow f=-1/2$$

Put the value of f in Eq. (v), we get,

$$g=-1$$

Also, put the values of f and g in Eq. (iv), we get

$$c=-8.$$

Hence, the equation of the circle is

$$x^2+y^2-2x-y-8=0$$

79. Equation of any circle passing through the point of intersection of the circles

$$x^2+y^2+13x-3y=0$$

and $2x^2+2y^2+4x-7y-25=0$

is $(x^2+y^2+13x-3y)$

$$+\lambda(2x^2+2y^2+4x-7y-25)=0 \quad \text{...(i)}$$

which is passing through (1, 1).

Therefore,
$(1+1+13-3)+\lambda(2+2+4-7-25)=0$
$\Rightarrow \quad 12-24\lambda=0$
$\Rightarrow \quad \lambda=\frac{1}{2}$
Put the value of λ in Eq. (i), we get
$(x^2+y^2+13x-3y)+\frac{1}{2}(2x^2+2y^2+4x-7y-25)=0$
$\Rightarrow (2x^2+2y^2+26x-6y)+(2x^2+2y^2+4x-7y-25)=0$
$\Rightarrow (4x^2+4y^2+30x-13y-25)=0$

80. The equation of radical axis is
$$(x^2+y^2+2x+3y+1)-(x^2+y^2+4x+3y+2)=0$$
$\Rightarrow \quad (2x+3y+1)-(4x+3y+2)=0$
$\Rightarrow \quad 2x+1=0$
$\Rightarrow \quad x=-\frac{1}{2}$
Thus, the equation of the circle is
$$(x^2+y^2+2x+3y+1)+\lambda(2x+1)=0$$
$\Rightarrow \quad (x^2+y^2+2(\lambda+1)x+3y+(1+\lambda))=0$
Since AB is diameter of the circle, so the centre lies on it.
Therefore, $-2\lambda-2+1=0$
$\Rightarrow \quad \lambda=-\frac{1}{2}.$
Hence, the equation of the circle is
$$(x^2+y^2+2x+3y+1)-\frac{1}{2}(2x+1)=0$$
$\Rightarrow \quad (x^2+y^2+2x+6y+1)=0$

81. Any circle passing through the point of intersection of the given line and the circle is
$$x^2+y^2-9+\lambda(x+y-1)=0$$
$\Rightarrow \quad x^2+y^2+\lambda x+\lambda y-(9+\lambda)=0$
So the centre is $\left(-\frac{\lambda}{2},-\frac{\lambda}{2}\right)$.
Since, the circle is smallest, so the centre $\left(-\frac{\lambda}{2},-\frac{\lambda}{2}\right)$ lies on the chord $x+y=1$.
Therefore, $-\frac{\lambda}{2}-\frac{\lambda}{2}=1 \Rightarrow \lambda=-1.$
Hence, the equation of the smallest circle is
$$x^2+y^2-9-(x+y-1)=0$$
$\Rightarrow \quad x^2+y^2-x-y-8=0$

82. The equation of any circle passing through the point of intersection of the given circle is
$(x^2+y^2-6x+2y+4)+\lambda(x^2+y^2+2x-4y-6)=0$
$\Rightarrow \quad (1+\lambda)x^2+(1+\lambda)y^2+2(\lambda-3)x+2(1-2\lambda)y+2(2-3\lambda)=0$
$\Rightarrow \quad x^2+y^2+2\left(\frac{\lambda-3}{\lambda+1}\right)x+2\left(\frac{1-2\lambda}{\lambda+1}\right)y+\left(\frac{2(2-3\lambda)}{\lambda+1}\right)=0$
So its centre is $\left(\frac{3-\lambda}{\lambda+1},\frac{2\lambda-1}{\lambda+1}\right)$.
Since, the centre lies on the line $y=x$,
$$\frac{2\lambda-1}{\lambda+1}=\frac{3-\lambda}{\lambda+1}$$
$\Rightarrow \quad 3\lambda=4$
$\Rightarrow \quad \lambda=4/3$
Hence, the equation of the circle is $x^2+y^2-\frac{10}{7}x-\frac{10}{7}y-\frac{12}{7}=0.$

83. The given circles are $x^2+y^2+2x+3y+1=0$ and $x^2+y^2+4x+3y+2=0$.
Hence, the equation of the common chord is $2x+1=0$.
Therefore, the equation of the circle is
$$(x^2+y^2+2x+3y+1)+\lambda(x^2+y^2+4x+3y+2)=0 \quad \ldots(i)$$
$\Rightarrow \quad (1+\lambda)x^2+(1+\lambda)y^2+2(1+2\lambda)x+3(1+\lambda)y+(1+2\lambda)=0$
$\Rightarrow \quad x^2+y^2+2\left(\frac{1+2\lambda}{(1+\lambda)}\right)x+3y+\frac{(1+2\lambda)}{(1+\lambda)}=0$
So, its centre is $\left(-\frac{1+2\lambda}{\lambda+1},-\frac{3}{2}\right)$
Since $2x+1=0$ is the diameter, so centre lies on it.
Therefore,
$$2\left(-\frac{1+2\lambda}{\lambda+1}\right)+1=0$$
$\Rightarrow \quad -2-4\lambda+\lambda+1=0$
$\Rightarrow \quad \lambda=-\frac{1}{3}$
Hence, the equation of the circle (i) is
$$2(x^2+y^2)+2x+6y+1=0$$